

There are 150 points in the exam, across 12 numbered problems, some with multiple parts. The total number of points for each problem is indicated. Each part of a problem is worth the same amount. There is also a bonus problem on the last page of the exam.

1. (24 points) Give an example of each of the following, or explain why it is not possible.

a) A matrix A that satisfies the polynomial equation $x^5 - x = 0$ that is neither a zero matrix nor an identity matrix.

$x^5 - x = x(x^4 - 1)$, so any matrix satisfying $A^4 = I$ satisfies the equation. And just have to ensure A is not zero or I .
eg. $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, (i) , $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

b) A vector in $\mathbb{R}^3$ that is not contained in the set $\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} \right\}$.

$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix}$ or $\begin{pmatrix} 21 \\ 6 \\ 19 \end{pmatrix}$ or ...

note the difference between a set and its span.

c) An operator A on $\mathbb{R}^3$ and a non-zero vector $\mathbf{v}$ so that $\mathbf{v}$ is in both the column space and the nullspace of A .

many examples are possible.

eg. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, or $\begin{pmatrix} 4 & 0 & -8 \\ 0 & 0 & 0 \\ 2 & 0 & -4 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}$

d) A product of two nilpotent matrices that is not nilpotent.

eg. $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

$\uparrow \quad \uparrow$
 nilpotent

$\nwarrow$ not nilpotent,
 since it has a non-zero
 eigenvalue (1).

e) A power $n \geq 1$ of $A = \begin{pmatrix} 0 & -2 \\ 1 & 1 \end{pmatrix}$ so that $A^n = I$.

not possible. $\det A = 2$, so $\det A^n = 2^n$

but $\det I = 1$

and so $A^n \neq I$ for any $n \geq 1$.

f) A matrix with trace 4 and no real eigenvalues.

Since $\lambda = \frac{\text{tr} A}{2} \pm \sqrt{\left(\frac{\text{tr} A}{2}\right)^2 - \det A}$, (for 2×2 matrices)
 it suffices to take $\text{tr} A = 4$ and $\left(\frac{\text{tr} A}{2}\right)^2 - \det A < 0$

$\Rightarrow \text{tr} A = 4$ and $4 < \det A$

Alternatively: $\begin{pmatrix} 2+i & 0 \\ 0 & 2-i \end{pmatrix}$ has $\text{tr} = 4$
 and no real eigenvalues

so eg. $\begin{pmatrix} 2 & 10 \\ -1 & 2 \end{pmatrix}$ or $\begin{pmatrix} 3 & -1000 \\ 200 & 1 \end{pmatrix}$

g) A nilpotent matrix with characteristic polynomial $\lambda^3 - 4\lambda^2 + \lambda + 6$.

not possible. A matrix with this characteristic polynomial has determinant -6, but a nilpotent matrix has determinant 0.

h) A matrix that has rank 3 and that has a left nullspace of dimension 4.

eg. $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, since A has rank 3 and
 since A^T has nullity 4.

2. (6 points) Give a precise definition for each of the following linear algebra terms.

nullspace:

The nullspace of a linear transformation $L: V \rightarrow W$ is the subset of V sent to the zero vector in W by L .

orthonormal:

In an inner product space, a set of vectors is orthonormal if each vector has norm 1 and each pair of distinct vectors is orthogonal (so that the inner product of the pair is 0.)

3. (9 points) Complete the following:

a) State the Spectral Theorem. Give an example that illustrates this result.

Let $A = A^*$ be a self-adjoint operator in an inner product space X . Then all eigenvalues of A are real, and there exists an orthonormal basis of eigenvectors of A in X . In the real case, this says that symmetric matrices are diagonalizable with an orthonormal change of basis matrix.

$$\text{E.g. } \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}^{-1}$$

b) State the QR decomposition theorem. Give an example of a matrix A that has a QR decomposition and a matrix B that does not have a QR decomposition. (You do not need to provide the decomposition.)

Let A be an invertible matrix in an inner product space. Then $A = QR$ where Q has orthonormal columns and R is upper triangular with positive diagonal entries.

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \text{ (b/c invertible)} \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \text{ (b/c not invertible)}$$

c) In some combination, clearly state three ways in which pairs of the following terms are related. You may use a term more than once:

orthogonal, trivial, nullspace, invertible, row space, column space, transpose, rank.

Many examples are possible. For instance:

The column space of the transpose of a matrix is its row space.

The nullspace of an invertible matrix is trivial.

The row space of a matrix is orthogonal to its nullspace.

The rank of a matrix is the dimension of its column space.

The inverse of an orthogonal matrix is its transpose.

4. (16 points) Prove **two** of the following:

- a) If two square matrices A and B are similar, then they have the same nullity.
- b) Suppose that a square matrix A has two eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ with corresponding eigenvalues λ_1 and λ_2 , such that $\mathbf{v}_1 + \mathbf{v}_2 \neq \mathbf{0}$. Show that $\mathbf{v}_1 + \mathbf{v}_2$ is an eigenvector of A if and only if $\lambda_1 = \lambda_2$.
- c) Let A be a real $m \times n$ matrix. Show that the rank of A is equal to the rank of $A^T A$.
- d) If the set of vectors $\{u, v, w\}$ is orthonormal, then this set of vectors is linearly independent.

First solution:

Various

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4. (continued) **Second solution:**

Various

5. (30 points) Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

a) Two orthogonal 3×3 matrices are similar to each other.

sometimes.

True example:

A ^{square} matrix is always similar to itself,

so

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

False example:

The only matrix that is similar to the identity matrix is itself,

so

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \not\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

b) The sum of two orthogonal matrices is invertible.

sometimes

True example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

↑ invertible

as

$$\begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

False example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

↑ not invertible.

c) A permutation matrix has 1 as one of its eigenvalues.

Always. Since the columns of a permutation matrix are just the standard basis vectors in some order, we have that the "all ones" vector $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is always an eigenvector, and with eigenvalue 1.

d) A given complex square matrix is either diagonalizable or invertible (or both).

sometimes

True example

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is both diagonalizable
& invertible

False example

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

is neither.

e) If A and B are real matrices and A is 3×2 and B is 2×3 , then BA has a left inverse.

sometimes

$$BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \leftarrow \text{invertible}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

True example:

$$\mathbb{F}^2 \xrightarrow{A} \mathbb{F}^3$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

False example:

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\mathbb{F}^3 \xrightarrow{B} \mathbb{F}^2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

composition
is injective,
so $\exists$
left inverse.

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

composition
is not
injective,
so $\nexists$ left
inverse.

f) Given an inner product on $\mathbb{R}^2$, the inner product of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is 0.

sometimes.

True example: the standard inner product
yields 0.

$$\text{False example: } (1 \ 0) \begin{pmatrix} 20 & 10 \\ 10 & 20 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= (1 \ 0) \begin{pmatrix} 10 \\ 20 \end{pmatrix} = 10$$

6. (6 points) Complete the following.

Suppose that for a certain norm on $\mathbb{R}^4$, the vector $\begin{pmatrix} 1 \\ 0 \\ 8 \\ 4 \end{pmatrix}$ has norm 8.

- a) Using this information about the norm, compute the norm of $\begin{pmatrix} -1 \\ 0 \\ -8 \\ -4 \end{pmatrix}$ and of $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$; or explain why you do not have enough information to do so.
- b) Give an example of a norm on $\mathbb{R}^4$ that has the above property.
- c) Explain how your example norm satisfies the defining properties of a norm.

a) norm of $\begin{pmatrix} -1 \\ 0 \\ -8 \\ 4 \end{pmatrix}$ is also 8, by homogeneity
 norm of $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ is 0, by non-degeneracy

b) for instance, the max norm (L^∞)
 or the standard norm, rescaled by $\frac{8}{9}$
 (or any other norm, appropriately rescaled)

c) various

7. (12 points)

a) Circle two matrices that are self-adjoint. *equal to its own conjugate transpose.*

$$\begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix} \quad \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} \quad \begin{pmatrix} 4i & 0 \\ 0 & 3i \end{pmatrix} \quad \begin{pmatrix} 4 & 1 \\ -1 & 3 \end{pmatrix} \quad \begin{pmatrix} 3 & 0 & -2-i \\ 0 & 2 & 0 \\ -2+i & 0 & 4 \end{pmatrix}$$

b) Circle two matrices that are not diagonalizable over $\mathbb{R}$.

$$\begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix} \quad \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 5 & 1 \\ 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

c) How many real 4×3 matrices are there that are in RREF and so that every entry is either a 1 or a 0?

$$\begin{pmatrix} - & - & - \\ - & - & - \\ - & - & - \\ - & - & - \end{pmatrix}$$

rank 0:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

rank 1:

$$\begin{pmatrix} - & - & - \\ \bigcirc & \bigcirc & \bigcirc \\ - & - & - \\ - & - & - \end{pmatrix}$$

1
 $2^3 + 2^1 + 2^0$
 $= 2^3 - 1$
 $= 7$

rank 2:

second row looks like

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} - & - & - \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

2 options for first row
 $\times 2$ options for second row = 4

3 options for first row

rank 3:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{only example.}$$

So total is
 $1 + 7 + 7 + 1 = 16$

8. (8 points)

Complete **one** of the following.

a) The $n \times n$ matrix S_n is a subtraction table mod n if the element of S_n in the i th row and j th column a_{ij} is equal to $(i - 1) - (j - 1) \pmod n$. For example, S_4 looks like this:

$$S_4 = \begin{pmatrix} 0 & 3 & 2 & 1 \\ 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \end{pmatrix}$$

Show that the determinant of S_n is always equal to 0 mod n .

b) Let the $n \times n$ matrix W_n be defined as having entries $a_{ij} = 2^{(i-j)}$. That is, the entry in the i th row and j th column is $2^{(i-j)}$. For all $n \geq 1$, find the determinant and the eigenvalues (with multiplicity) of W_n .

Ideas: (mod n)

a) Through column operations, one can find a linear dependence among the columns. Or instead, can make a column so that all entries are a multiple of n , and then apply cofactor expansion

b) Each column is a scalar multiple of the first. So for $n > 1$, we have $\det(W_n) = 0$. Further analysis yields eigenvalues of 0 (w/alg mult $n-1$) and n (w/alg mult 1).

9. (9 points) Complete the following.

a) Compute the characteristic polynomial and the minimal polynomial of the following matrix M . Are they equal to each other?

$$M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{pmatrix}$$

b) There are 9 matrices that can be formed by changing exactly one entry of M to a 2. Give an example of such a matrix that has the same characteristic polynomial as M and the same minimal polynomial as M .

c) Again, there are 9 matrices that can be formed by changing exactly one entry of M to a 2. Give an example of such a matrix that has the same characteristic polynomial as M but a *different* minimal polynomial than M .

a) char poly: $(4-\lambda)^3$ min poly: $(4-\lambda)^2$

b) eg. $\begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 4 \end{pmatrix}$

c) eg. $\begin{pmatrix} 4 & 2 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{pmatrix}$ with min poly $(4-\lambda)^3$

10. (4 points)

For **one** of the following matrices, compute its 100th power. You should write your answer as a single matrix, but its entries can be left as computations, such as $4 \cdot 5^{100} - 2 \cdot 5^{50}$.

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

$$C = \begin{pmatrix} 5 & 4 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

A is a permutation matrix with the following cycle structure: $e_1 \rightarrow e_3 \rightarrow e_5$ and $e_2 \rightarrow e_4 \rightarrow e_6$

and so after 99 applications of A we get the identity, and after 100, we just get A. $A^{100} = A$.

B is diagonalizable. $\lambda = 2$ with eigenvector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$,

$\lambda = -1$ with eigenvector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$,

and $\lambda = 6$ with eigenvector $\text{Nul} \begin{pmatrix} -4 & 0 & 1 \\ 0 & -7 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$

$$\text{So } B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}^{-1}$$

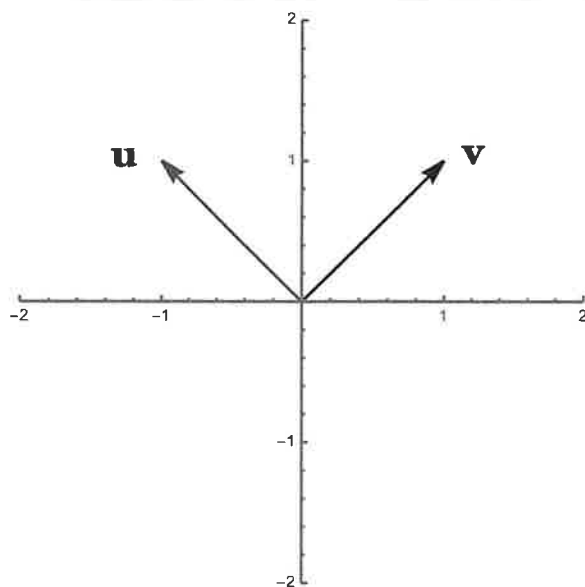
$$\text{and } B^{100} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2^{100} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6^{100} \end{pmatrix} \begin{pmatrix} 1 & 0 & -1/4 \\ 0 & 1 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -4 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 1 & 0 & -1/4 \\ 0 & 1 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{100} & 0 & -\frac{2^{100}}{4} + \frac{6^{100}}{4} \\ 0 & 1 & 0 \\ 0 & 0 & 6^{100} \end{pmatrix}$$

$$C^{100} = \begin{pmatrix} 5^{100} & 4 \cdot 5^{99} & 3 \cdot 5^{99} & 2 \cdot 5^{99} & 5^{99} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$



11. (6 points)

a) Write the matrix representing the following linear transformation of $\mathbb{R}^2$ using the standard basis:

scale $\mathbf{u}$ by 6 and $\mathbf{v}$ by 8

b) Then compute the singular values of the square of this matrix.

$$\begin{aligned}
 A &= \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 8 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \\
 &= \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 8/\sqrt{2} & 8/\sqrt{2} \\ -6/\sqrt{2} & 6/\sqrt{2} \end{pmatrix} \\
 &= \begin{pmatrix} 7 & 1 \\ 1 & 7 \end{pmatrix}
 \end{aligned}$$

This can also be obtained by squaring A , seeing that its eigenvalues are 64 and 36, and seeing that the eigenvalues of $(A^T A)^2$ are $36^2 + 64^2$, and so the s.v. of A^2 are 36 and 64.

Since A is orthogonally diagonalized above, this is the SVD of A . So the SVD of A^2 has "diagonal" matrix (which in this case is actually diagonal) $\begin{pmatrix} 64 & 0 \\ 0 & 36 \end{pmatrix}$. So the singular values of A^2 are 64 and 36.

12. (20 points)

Complete the following. **Use the next four blank pages, as needed.** Label each part of your solution clearly, (a) through (j). (These need not be completed in order.)

Let A be the real matrix $\begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

a) Compute the rank and determinant of A .

*rank = 4, det = (3)(-1) = -3
since A is block diagonal.*

b) Compute the characteristic polynomial and eigenvalues of A .

c) Compute bases for the eigenspaces of A .

d) Give the general solution to the matrix equation $Ax = \begin{pmatrix} 4 \\ -1 \\ 0 \\ 0 \end{pmatrix}$.

e) Compute the result of applying A^{99} to the vector $\begin{pmatrix} -1 \\ 1 \\ -3 \\ 3 \end{pmatrix}$.

f) Compute the inverse of A ; or explain why this is not possible

g) Give the Jordan canonical form of A ; or explain why A cannot be put in this form.

h) Give the minimal polynomial of A .

i) Do one of the following: (i) Give a QR decomposition of A ; or explain why this is not possible. (ii) Write A as a product of elementary matrices, or explain why this is not possible.

j) Circle the terms that apply to A , and do not circle the ones that don't:

normal invertible unitary nilpotent diagonalizable orthogonal symmetric

Solutions for problem 12: Rank=4

a) since A is block diagonal, we have $\det A = (3)(-1) = -3$.
other approaches are possible, eg. cofactor expansion

b) char poly $A = \det \begin{pmatrix} 2-\lambda & 1 & 0 & 0 \\ 1 & 2-\lambda & 0 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 1 & \lambda \end{pmatrix}$

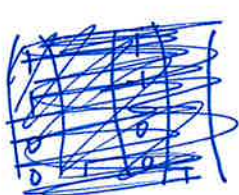
Again, $A - \lambda I$ is block diagonal, so we have

$$[(2-\lambda)(2-\lambda) - 1][\lambda^2 - 1]$$

$$= (\lambda^2 - 4\lambda + 3)(\lambda^2 - 1) = (\lambda - 3)(\lambda - 1)(\lambda - 1)(\lambda + 1)$$

so $\lambda = 3, -1, 1, 1$ ↖ multiplicity 2.

c) with corresponding eigenvectors



$$\lambda = 3: \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \lambda = -1: \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\lambda = 1: \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

d) $\left(\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 4 \\ 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array} \right) \xrightarrow{R3 \leftrightarrow R4} \left(\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 4 \\ 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$

$\xrightarrow{\text{RREF}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right)$ so $\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 0 \\ 0 \end{pmatrix}$

Solutions for problem 12:

e) Since $\begin{pmatrix} -1 \\ 1 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -3 \\ 3 \end{pmatrix}$ and each of those

is an eigenvector, we have

$$A^{99} \begin{pmatrix} -1 \\ 1 \\ -3 \\ 3 \end{pmatrix} = (1)^{99} \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + (-1)^{99} \begin{pmatrix} 0 \\ 0 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 3 \\ -3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 3 \\ -3 \end{pmatrix}$$

f) We can compute this in blocks, for instance.

$$A^{-1} = \left(\begin{array}{cc|cc} 2/3 & -1/3 & 0 & 0 \\ -1/3 & 2/3 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

g) Since $AM=GM$ for all eigenvalues of A , A is diagonalizable and so its JCF is

$$\begin{pmatrix} 3 & & 0 \\ & 1 & \\ 0 & & -1 \end{pmatrix}$$

h) The minimal polynomial of A divides its characteristic polynomial, and so equals

$$p = (\lambda - 3)(\lambda + 1)(\lambda - 1)^2 \text{ or } q = (\lambda - 3)(\lambda + 1)(\lambda - 1).$$

And since A is diagonalizable (and similar matrices have the same min/char poly) we have that the min poly of A is q .

Solutions for problem 12:

i.i)

$$A = \begin{pmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} & 0 & 0 \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{5} & \frac{4\sqrt{5}}{5} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

i.ii)

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 0 & 0 & | & 1 \\ 1 & 2 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & 0 & | & 1 \end{pmatrix}$$

$$R_1 + R_2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$R_3 \leftrightarrow R_4 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$R_1 - R_2$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$R_2 - R_1$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$\frac{1}{3}R_2$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

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Solutions for problem 12: